

Notes: Ferreira, Manso, and Silva (RFS, 2014)

Incentives to Innovate and the Decision to Go Public or Private

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1 Big picture

- **Question.** How does a firm's ownership structure — **public** versus **private** — shape managers' incentives to invest in innovative projects, and when is it optimal to go public or private?
- **Setting.** The paper develops a **two-period theoretical model** in which an insider-manager chooses between a **conventional project** (exploitation of existing ideas) and an **innovative project** (exploration of new ideas), while also choosing whether the firm is financed and owned through **public markets** or **private investors**.
- **Core challenge.** Public and private ownership differ along many dimensions: transparency, liquidity, capital-raising costs, monitoring, and market pricing. The paper isolates one central difference — **public firms are more transparent than private firms** — and studies how that difference changes innovation incentives.
- **Main contribution.** The paper shows that:
 1. **private ownership** can create a **tolerance-for-failure effect**, because insiders can sell after bad news when outsiders cannot fully see the failure;
 2. **public ownership** can create **short-termism**, because market prices react quickly to good news, making it attractive to choose safer projects that look good early on;
 3. these two mechanisms imply systematic differences in innovation and in the optimal decision to go public or private.
- **Why it matters.** A large literature argues that public markets create short-term pressure. This paper goes further by showing that *private ownership is not just “less short-term”*; it can positively encourage experimentation because failure is easier to absorb when information is opaque and insiders can time the market.
- **Main mechanism.**
 - Under **private ownership**, outsiders do *not* observe interim earnings x_1 . If the innovative project fails early, the insider may still sell shares at a price that pools failure with liquidity-driven sales. That option makes early failure less painful and encourages innovation.
 - Under **public ownership**, interim earnings are observable. A failure cannot be disguised, so there is *no* tolerance for failure. At the same time, if the insider chooses the conventional project and gets early success, the stock market reacts quickly and allows the insider to cash out. That encourages conventional, early-success projects.
- **Main theoretical results.**
 - The equilibrium intensity of innovation is always at least as high under private ownership as under public ownership: $\sigma_{\text{priv}}^* \geq \sigma_{\text{pub}}^*$.
 - If innovation is socially efficient and public/private capital costs are the same, the firm optimally prefers **private ownership**.
 - If the conventional project is efficient, the firm prefers **public ownership** unless private and public are tied and both implement the conventional project.
 - When private securities are illiquid, public ownership can become preferable even when innovation is efficient, because private ownership then trades off better innovation incentives against worse liquidity.

- **Life-cycle interpretation.** The paper interprets ownership choice as a **life-cycle problem**: firms should be private when they need exploration, experimentation, or restructuring, and should go public when they are ready to exploit proven ideas.
- **Economic takeaway.** Private firms are biased toward **exploration**; public firms are biased toward **exploitation**. The decision to go public or private depends on which type of activity is more valuable at that stage of the firm’s life.

2 Data and measurement (key objects)

2.1 This is a theory paper: the “data” are model primitives

- There is **no paper-specific estimation dataset**. The paper is a pure theory paper.
- The key objects are the model’s primitives: project payoffs, liquidity shocks, investor types, transparency, issuance costs, and the ownership choice.
- The paper ends with a set of **empirical predictions** and discusses related evidence from other studies, but it does not estimate the model structurally or run a paper-specific reduced-form exercise.

2.2 Projects, payoffs, and innovation

- The insider chooses between two projects at date 0 and again at date 1:
 - **Project 1:** a conventional project, interpreted as **exploitation of existing ideas**;
 - **Project 2:** an innovative project, interpreted as **exploration of new ideas**.
- Each project produces earnings over **two consecutive periods**. In each period, earnings are either **success** or **failure**.
- The conventional project succeeds with publicly known probability p .
- The innovative project’s success probability is initially unknown. Its expected success probability is revised after observing the first-period outcome:
 - after early failure, beliefs about future success fall;
 - after early success, beliefs about future success rise.
- The paper parameterizes the innovative project so that

$$\delta p = E[q], \quad \theta p = E[q \mid S],$$

with $0 < \delta < 1$ and $\theta > 1$.

- **Interpretation.** The innovative project is less likely to succeed early than the conventional project, but if it succeeds early, it becomes much more promising later on. So it is a natural model of experimentation.

2.3 Project values and the exploration–exploitation trade-off

- Normalizing failure payoff to 0 and success payoff to 1, the ex ante value of the conventional project is

$$v_1 = 2p.$$

- If the insider starts with the innovative project and switches back to the conventional project after early failure, the ex ante value is

$$v_2 = p\{1 + \delta[1 + p(\theta - 1)]\}.$$

- Innovation is ex ante efficient if and only if

$$v_2 - v_1 > 0 \iff \delta[1 + p(\theta - 1)] > 1.$$

- **Interpretation.** This is the central technological trade-off: project 2 has lower early success but larger upside if it works.

2.4 Liquidity shocks and preferences

- The insider is risk neutral but may face a **liquidity shock** at date 1.
- Preferences are reduced-form Diamond–Dybvig style:

$$U(c_1, c_2) = \begin{cases} c_1 & \text{with probability } \mu, \\ c_2 & \text{with probability } 1 - \mu. \end{cases}$$

- With probability μ , the insider becomes an **early consumer** and wants liquidity at date 1.
- With probability $1 - \mu$, the insider is a **late consumer** and can wait until date 2.
- **Interpretation.** Liquidity shocks are what create a motive to sell shares early. Without such shocks, insider selling would always look purely strategic.

2.5 Investor types and market access

- There are two investor types:
 - **unsophisticated investors**, who only observe public information and price securities competitively;
 - **sophisticated investors**, who observe the same information as insiders when they trade.
- Sophisticated investors are in short supply. The insider can find one only with probability $e < 1$.
- If the insider meets a sophisticated investor, the trade can be based on fundamentals; otherwise, the insider must sell to public investors at a market price based only on public information.
- **Interpretation.** Parameter e captures how easy it is to obtain informed private liquidity, for example through venture capital or private equity.

2.6 Public versus private ownership structures

- The paper's key structural difference between ownership forms is **transparency of interim earnings**:
 - under **public ownership**, interim earnings x_1 are observed by everyone;
 - under **private ownership**, only the insider, current private investors, and future sophisticated investors observe x_1 .
- The model also allows two additional differences:
 - **capital-raising costs**: each dollar sold yields only c_{pub} under public ownership and only c_{priv} under private ownership;
 - **liquidity of insider shares**: if the insider sells private shares at date 1, each dollar sold may yield only $k \leq 1$, whereas public shares have no such liquidity discount.

2.7 Timing

- **Date 0:**
 1. choose ownership structure, public or private;
 2. sell an initial fraction of shares to raise the investment cost I ;
 3. choose the project, conventional or innovative.
- **Date 1:**
 1. interim earnings x_1 are realized;
 2. the insider may switch projects;
 3. the insider learns whether there is a liquidity shock;
 4. the insider decides whether to sell shares.
- **Date 2:**
 1. second-period earnings x_2 are realized;
 2. shareholders receive dividends;
 3. the firm is liquidated.

2.8 Key frictions and intuition

- **Borrowing constraints** make liquidity shocks matter.
- **Opacity under private ownership** allows insiders to pool bad-news sales with liquidity-driven sales.
- **Transparency under public ownership** makes bad news observable, but also makes good-news cash-outs possible.
- **Resulting intuition:**

- private ownership is more **failure-tolerant**;
- public ownership is more **short-termist**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Technology and project values

With failure normalized to 0 and success normalized to 1, the conventional project has value

$$v_1 = 2p.$$

The innovative project, with the option to switch back to the conventional project after early failure, has value

$$v_2 = p\{1 + \delta[1 + p(\theta - 1)]\}.$$

Innovation is efficient if

$$v_2 - v_1 > 0.$$

Interpretation.

- Project 1 is safer in the short run.
- Project 2 is riskier initially but can generate a much better continuation value if the first experiment succeeds.
- This creates a clean exploration–exploitation problem.

3.2 Private ownership: sale decisions and market beliefs

Under private ownership, outsiders do not observe x_1 . Let $b \in [0, 1]$ denote the probability that an insider *without* liquidity needs sells shares to the market after observing failure $x_1 = 0$.

The market's posterior belief that a public sale is driven by liquidity needs is

$$m(\sigma_{\text{priv}}, b) = \frac{\Pr(\text{Sale} \mid \text{Liquidity needs}) \Pr(\text{Liquidity needs})}{\Pr(\text{Sale})}.$$

Using Bayes' rule, this becomes

$$m(\sigma_{\text{priv}}, b) = \frac{\mu(1 - e)}{\mu(1 - e) + b(1 - \mu + \mu e)[\sigma_{\text{priv}}(1 - \delta p) + (1 - \sigma_{\text{priv}})(1 - p)]}.$$

The market value of shares in a private sale to the public is

$$V_{\text{priv}}(\sigma_{\text{priv}}, b) = m(\sigma_{\text{priv}}, b)[\sigma_{\text{priv}}v_2 + (1 - \sigma_{\text{priv}})v_1] + (1 - m(\sigma_{\text{priv}}, b))p.$$

The paper then shows two key lemmas:

- after **success**, a late consumer never sells to the public market;
- after **failure**, a late consumer sells with probability one in the baseline private case.

Interpretation.

- If outsiders cannot observe x_1 , they cannot tell whether a sale follows bad news or a liquidity shock.
- That makes it possible to sell after failure at a price above the post-failure fundamental value p .
- This embedded exit option is the source of **tolerance for failure**.

3.3 Private ownership: incentive compatibility and innovation

Define the private-case option value of exiting early after failure as

$$T(\sigma_{\text{priv}}) \equiv V_{\text{priv}}(\sigma_{\text{priv}}, 1) - p.$$

The incentive compatibility condition for choosing the innovative project is

$$u_{\text{priv},2} \geq u_{\text{priv},1} \iff v_2 - v_1 + p(1 - \delta)T(\sigma_{\text{priv}}) \geq 0.$$

The term $v_2 - v_1$ is the **technological benefit** of innovation. The term

$$p(1 - \delta)T(\sigma_{\text{priv}})$$

is the **pecuniary benefit** of innovation coming from the option to sell after bad news.

Interpretation.

- Even if the innovative project is technologically inferior, the insider may still choose it because failure creates a valuable trading opportunity.
- This is the paper's central **tolerance-for-failure effect**.

3.4 Equilibrium innovation under private ownership

Proposition 1. There is a unique equilibrium exploration probability $\sigma_{\text{priv}}^* \in [0, 1]$ such that:

- if $v_2 \geq v_1$, then $\sigma_{\text{priv}}^* = 1$;
- if $v_2 < v_1$, then

$$\sigma_{\text{priv}}^* = \begin{cases} 1, & \text{if } \frac{v_1 - v_2}{p(1 - \delta)} \leq T(1), \\ \sigma \text{ such that } T(\sigma) = \frac{v_1 - v_2}{p(1 - \delta)}, & \text{if } T(1) < \frac{v_1 - v_2}{p(1 - \delta)} < T(0), \\ 0, & \text{if } T(0) \leq \frac{v_1 - v_2}{p(1 - \delta)}. \end{cases}$$

Interpretation.

- If innovation is efficient, the private firm always innovates.
- If innovation is inefficient, the private firm may still innovate with positive probability, or even with certainty, because failure is more tolerable under opacity.
- So private ownership is biased **toward innovation**.

Proposition 2. In the interior case $\sigma_{\text{priv}}^* \in (0, 1)$,

$$\frac{\partial \sigma_{\text{priv}}^*}{\partial \theta} > 0, \quad \frac{\partial \sigma_{\text{priv}}^*}{\partial \delta} > 0, \quad \frac{\partial \sigma_{\text{priv}}^*}{\partial \mu} > 0, \quad \frac{\partial \sigma_{\text{priv}}^*}{\partial e} < 0.$$

Interpretation.

- A higher θ or δ makes innovation technologically better, so innovation rises.
- A higher μ makes sales more likely to be interpreted as liquidity-driven, which strengthens tolerance for failure.
- A higher e makes informed private liquidity easier to find, which reduces the need to exploit uninformed public markets after failure and weakens the private bias toward innovation.

3.5 Value of private ownership

The expected ex ante value per retained share under private ownership equals

$$u_{\text{priv}} = \sigma_{\text{priv}}^* v_2 + (1 - \sigma_{\text{priv}}^*) v_1.$$

The insider sells just enough shares to finance the initial investment:

$$\alpha_{\text{priv}}^* = 1 - \frac{I}{c_{\text{priv}} [\sigma_{\text{priv}}^* v_2 + (1 - \sigma_{\text{priv}}^*) v_1]}.$$

Hence the insider's value of the private firm is

$$W_{\text{priv}} = \sigma_{\text{priv}}^* v_2 + (1 - \sigma_{\text{priv}}^*) v_1 - \frac{I}{c_{\text{priv}}}.$$

Interpretation.

- The insider internalizes the project's value net of the cost of raising private capital.
- In the baseline model, strategic trading at date 1 does not create ex ante rents on average; prices at date 0 adjust so investors break even.
- The inefficiency of the private firm comes from *distorted project choice*, not from a free lunch in security trading.

3.6 Public ownership: selling after success and market pricing

Under public ownership, interim earnings x_1 are publicly observed. If the market observes a success and a sale, it assigns probability

$$s \equiv \Pr(\text{Project 2} \mid \text{Sale}, x_1 = 1) = \frac{\Pr(\text{Sale}, x_1 = 1 \mid \text{Project 2}) \Pr(\text{Project 2})}{\Pr(\text{Sale}, x_1 = 1)}.$$

In equilibrium this becomes

$$s(\sigma_{\text{pub}}) = \frac{\sigma_{\text{pub}}\mu(1-e)\delta}{1 - \sigma_{\text{pub}} + \sigma_{\text{pub}}\mu(1-e)\delta}.$$

The market value of shares after a successful public sale is

$$V_{\text{pub}}(\sigma_{\text{pub}}) = 1 + s(\sigma_{\text{pub}})\theta p + (1 - s(\sigma_{\text{pub}}))p = 1 + p + s(\sigma_{\text{pub}})p(\theta - 1).$$

The paper shows:

- if the insider chose the **innovative** project and has no liquidity needs, he never sells after success;
- if the insider chose the **conventional** project and succeeds, he weakly prefers to sell after success.

Interpretation.

- Public markets see success, but they cannot perfectly infer *which project* caused that success.
- An insider who played it safe can exploit that ambiguity by selling after a good early outcome.
- This is the paper's **short-termism** channel.

3.7 Equilibrium innovation under public ownership

The expected ex ante value per share if the insider chooses the conventional project is

$$u_{\text{pub},1} = pV_{\text{pub}}(\sigma_{\text{pub}}) + (1-p)p.$$

If the insider chooses the innovative project, the expected value is

$$u_{\text{pub},2} = \delta p \left[\mu(1-e)V_{\text{pub}}(\sigma_{\text{pub}}) + (1-\mu+\mu e)(1+\theta p) \right] + (1-\delta p)p.$$

Proposition 3. There is a unique equilibrium exploration probability $\sigma_{\text{pub}}^* \in [0, 1)$ given by

$$\sigma_{\text{pub}}^* = \frac{s^*}{\mu(1-e)\delta + s^*[1 - \delta\mu(1-e)]},$$

where

$$s^* = \max \left\{ \frac{v_2 - v_1 - \delta\mu(1-e)p^2(\theta - 1)}{p^2(\theta - 1)[1 - \delta\mu(1-e)]}, 0 \right\}.$$

Moreover,

- if $v_1 \geq v_2 - \delta\mu(1 - e)p^2(\theta - 1)$, then $\sigma_{\text{pub}}^* = 0$;
- if $v_1 < v_2 - \delta\mu(1 - e)p^2(\theta - 1)$, then $\sigma_{\text{pub}}^* \in (0, 1)$.

Interpretation.

- Under public ownership, full innovation never occurs in equilibrium.
- If the conventional project is efficient, public ownership always implements it.
- Even if innovation is efficient, the insider may still choose the conventional project with positive probability because it produces earlier success and earlier cash-out opportunities.

Proposition 4. In the interior case $\sigma_{\text{pub}}^* \in (0, 1)$,

$$\frac{\partial \sigma_{\text{pub}}^*}{\partial \delta} > 0, \quad \frac{\partial \sigma_{\text{pub}}^*}{\partial \theta} > 0, \quad \frac{\partial \sigma_{\text{pub}}^*}{\partial \mu} < 0, \quad \frac{\partial \sigma_{\text{pub}}^*}{\partial e} > 0.$$

Interpretation.

- Better innovative upside (δ or θ) promotes innovation, as usual.
- A higher μ makes sales more likely to be interpreted as liquidity-driven and therefore reduces the penalty for cashing out after early success under the conventional project. That strengthens short-termism and *reduces* innovation.
- A higher e gives insiders another way to obtain liquidity, softening public-market short-term pressure and increasing innovation.

3.8 Value of public ownership

The ex ante value per retained share under public ownership equals

$$u_{\text{pub}} = \sigma_{\text{pub}}^* v_2 + (1 - \sigma_{\text{pub}}^*) v_1.$$

The insider retains

$$\alpha_{\text{pub}}^* = 1 - \frac{I}{c_{\text{pub}} [\sigma_{\text{pub}}^* v_2 + (1 - \sigma_{\text{pub}}^*) v_1]}.$$

Hence the insider's value of the public firm is

$$W_{\text{pub}} = \sigma_{\text{pub}}^* v_2 + (1 - \sigma_{\text{pub}}^*) v_1 - \frac{I}{c_{\text{pub}}}.$$

Interpretation.

- The value of a public firm differs from the private case for two reasons: different innovation intensity and different capital-raising costs.
- Public ownership may be attractive even if it distorts innovation, simply because public capital is cheaper or public shares are more liquid.

3.9 The decision to go public or private

Define the relative cost advantage of public over private financing as

$$a \equiv \frac{1}{c_{\text{priv}}} - \frac{1}{c_{\text{pub}}} = \frac{c_{\text{pub}} - c_{\text{priv}}}{c_{\text{priv}}c_{\text{pub}}}.$$

Using the value expressions, the ownership choice is characterized by

$$W_{\text{priv}} - W_{\text{pub}} = (\sigma_{\text{priv}}^* - \sigma_{\text{pub}}^*)(v_2 - v_1) - aI.$$

Proposition 5. Private ownership is weakly preferable if and only if

$$(\sigma_{\text{priv}}^* - \sigma_{\text{pub}}^*)(v_2 - v_1) \geq aI.$$

Interpretation.

- The choice depends on three objects:
 1. how much more innovative private firms are than public firms,
 2. how valuable innovation is relative to conventional projects,
 3. how much cheaper public capital is.
- Private ownership wins when its innovation advantage outweighs the public capital-cost advantage.

Proposition 6.

$$\sigma_{\text{priv}}^* - \sigma_{\text{pub}}^* \geq 0$$

for all parameter values.

Interpretation.

- Private ownership is always at least as innovation-intensive as public ownership.
- The reason is asymmetric: the private structure may induce **too much** innovation, while the public structure may induce **too little** innovation.

If capital costs are equal ($a = 0$), Corollary 1 implies:

- if $v_2 > v_1$, the insider chooses **private ownership**;
- if $v_1 > v_2$, the insider chooses **public ownership** or is indifferent only when both structures implement the conventional project;
- if $v_1 = v_2$, the insider is indifferent.

3.10 Illiquid private securities

The paper next allows private shares sold at date 1 to be illiquid, so that each dollar sold yields only $k < 1$ to the insider.

With illiquid private shares, a late consumer who sees failure may sell with probability

$$b(\sigma_{\text{priv}}) = \begin{cases} 1, & \text{if } k \geq k_1, \\ \frac{\mu(k[v_1 + \sigma_{\text{priv}}(v_2 - v_1)] - p)}{(1-k)(1-\mu)[1-p + \sigma_{\text{priv}}p(1-\delta)]p}, & \text{if } k_2 < k < k_1, \\ 0, & \text{if } k \leq k_2, \end{cases}$$

with thresholds

$$k_1 \equiv \frac{\mu p + (1-\mu)[1-p + \sigma_{\text{priv}}p(1-\delta)]p}{\mu[v_1 + \sigma_{\text{priv}}(v_2 - v_1)] + (1-\mu)[1-p + \sigma_{\text{priv}}p(1-\delta)]p}, \quad k_2 \equiv \frac{p}{v_1 + \sigma_{\text{priv}}(v_2 - v_1)}.$$

The option value becomes

$$T(\sigma_{\text{priv}}) = \max\{kV_{\text{priv}}(\sigma_{\text{priv}}, b) - p, 0\}.$$

Interpretation.

- When private markets are very illiquid, the insider cannot profitably sell after failure, so tolerance for failure disappears.
- When private markets are sufficiently liquid, the original private-ownership mechanism survives.

Proposition 7.

- If $v_2 > v_1$, then $\sigma_{\text{priv}}^* = 1$.
- If $v_2 < v_1$, the private case again has a piecewise characterization like Proposition 1, but now based on the modified option value with $k < 1$.
- If $v_2 = v_1$, multiple probabilities may be consistent with equilibrium because the insider can be indifferent over a range of strategies.

The private-firm value becomes

$$W_{\text{priv}} = \sigma_{\text{priv}}^* v_2 + (1 - \sigma_{\text{priv}}^*) v_1 - \frac{I}{c_{\text{priv}}} - L(k),$$

where $L(k)$ is the expected illiquidity cost from having to sell private shares at a discount.

The ownership condition becomes

$$(\sigma_{\text{priv}}^* - \sigma_{\text{pub}}^*)(v_2 - v_1) \geq aI + L(k).$$

Proposition 8. If $c_{\text{priv}} = c_{\text{pub}}$ and $k \leq 1$,

- if $v_1 \geq v_2$, the insider chooses the **public** structure;
- if $v_2 > v_1$, there exists a unique threshold $k^* \in (0, 1)$ such that the insider chooses public ownership if $k < k^*$ and private ownership if $k \geq k^*$.

Interpretation.

- Private ownership may be better for innovation but worse for liquidity.
- So the development of IPO markets or secondary liquidity for private claims can itself foster innovation.

3.11 Model implications and testable predictions

The paper's final section translates the theory into seven empirical predictions:

1. firms undertake more innovative projects after going private;
2. firms undertake fewer innovative projects after going public;
3. firms should go or stay private when innovative projects have higher net present value, and go or stay public when conventional projects are better;
4. a reduction in IPO costs fosters innovation;
5. an active buyout market fosters innovation in public firms but harms innovation in private firms;
6. greater information asymmetry in IPO pricing fosters innovation in private firms;
7. lower stock liquidity fosters innovation in public firms.

Interpretation.

- These predictions give the paper its empirical bite even though the model itself is theoretical.
- The comparative statics line up cleanly with the two key forces: **tolerance for failure** on the private side and **short-termism** on the public side.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- A standard short-termism story explains why public markets may discourage innovation, but it does not by itself explain why **private ownership actively encourages** innovation.
- Standard going-public models often emphasize only capital costs, monitoring, control, or liquidity. This paper adds a distinct mechanism based on **who observes interim information** and **when insiders can trade**.

4.2 What pins down the mechanism in the model

- The model's key assumption is minimal and sharp: **public firms reveal x_1 broadly; private firms do not**.
- Liquidity shocks are needed so that early sales can sometimes be innocent rather than purely strategic.
- The innovative project must be characterized by **low early success but high conditional upside**, so that there is a meaningful exploration–exploitation trade-off.
- Sophisticated investors must be scarce enough that insiders cannot always get fully informed private liquidity. Otherwise, public-market timing opportunities would disappear.

4.3 Why the comparative statics are useful here

- Parameters μ , e , and k each map into a clear economic object:
 - μ : frequency of liquidity shocks / extent to which sales look nonstrategic,
 - e : availability of informed private capital,
 - k : liquidity of private securities.
- Because the signs of the comparative statics differ between the public and private cases, the model generates many empirical predictions, not just one broad claim that private firms innovate more.

4.4 Economic credibility and caveats

- The model deliberately strips away many real-world features to isolate the ownership mechanism:
 - no debt in the baseline model,
 - no direct governance or monitoring advantage of public ownership,
 - no explicit product-market competition,
 - only two dates of cash-flow realization,
 - a highly stylized representation of insider trading and private liquidity.
- The paper is therefore best read as a **mechanism paper**, not as a complete theory of the corporation.
- The authors are explicit that additional forces, such as the benefits of hands-off public shareholders, could offset or attenuate the effects they emphasize.

4.5 Relation to the literature

- **Managerial short-termism:** closely related to Stein (1989), but with an endogenous public/private ownership choice.
- **Information disclosure and innovation:** related to Bhattacharya and Ritter (1983) and Maksimovic and Pichler (2001), but here the main channel runs through insiders' incentives, not rival learning.
- **Insider trading and innovation:** related to Hirshleifer (1971) and Bebchuk and Fershtman (1994), because private trading opportunities can distort project choice.
- **Going public or private:** connected to Shah and Thakor (1988), Zingales (1995), Chemmanur and Fulghieri (1999), Boot, Gopalan, and Thakor (2006), but the novelty is the explicit role of innovation incentives.

5 Tables and figures

5.1 Figure 1: technology tree and the option to switch

- Figure 1 lays out the paper’s basic experimentation structure.
- If the insider starts with the conventional project, success and failure are straightforward and the project continues.
- If the insider starts with the innovative project, success leads to continuation with the innovative method, while failure leads to switching back to the conventional method.
- **Read:** The figure shows why project 2 is not simply “riskier”; it is **learnable**. Failure teaches the insider to abandon the new method, while success raises the conditional continuation value.

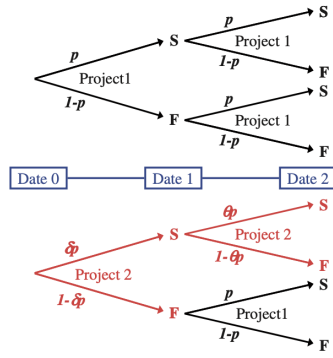


Figure 1
Earnings and probabilities associated with each initial project choice

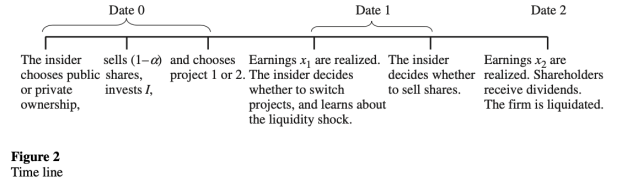


Figure 2
Time line

5.2 Figure 2: timeline and information flow

- Figure 2 shows the exact timing:
 1. date 0: ownership, financing, investment, and project choice;
 2. date 1: x_1 is realized, the insider may switch projects, a liquidity shock may arrive, and the insider decides whether to sell shares;
 3. date 2: final earnings are realized and the firm is liquidated.
- **Read:** The entire paper hinges on this timing. Ownership matters because the information available to outside investors at date 1 differs across public and private firms.

5.3 Figure 3: equilibrium innovation in the private case

- Figure 3 illustrates the private-equilibrium choice when the conventional project is technologically superior ($v_1 > v_2$).
- The vertical axis is the tolerance-for-failure term $T(\sigma_{\text{priv}})$ and the horizontal dashed lines represent different values of the scaled technological disadvantage of innovation.
- **Read:**

- if the technological disadvantage is large, the insider chooses the conventional project;
- if it is intermediate, the insider mixes;
- if it is small, the insider chooses the innovative project despite its lower technological value.
- This is the clearest visual summary of how private opacity can lead to **inefficiently high innovation**.

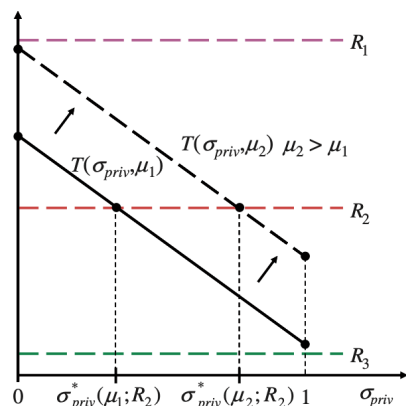


Figure 3
Equilibrium probability of innovation if exploitation is efficient

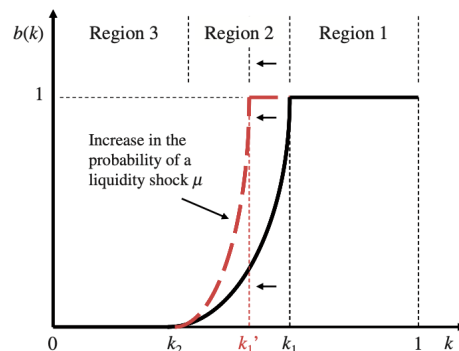


Figure 4
 $b(k)$: probability of a late-consumer insider selling shares after $x_1 = 0$

5.4 Figure 4: illiquid private securities and exit after failure

- Figure 4 shows how the probability of selling private shares after failure depends on private-market liquidity k .
- There are three regions:
 - if k is low, the insider never sells after failure;
 - for intermediate k , the insider mixes;
 - if k is high, the insider always sells after failure.
- **Read:** This figure shows exactly why private liquidity matters. The private ownership advantage is not automatic; it requires that insiders can actually monetize the option to exit after bad news.

5.5 Predictions section as the paper's empirical roadmap

- The paper has no formal estimation table, but its **Prediction 1–7** section functions like an empirical roadmap.
- The most important ones are:
 - public-to-private transitions should increase innovation;
 - going public should reduce innovation;
 - liquid public markets can worsen short-termism;
 - better private-market liquidity can make private ownership more attractive for innovative firms.

- The paper then discusses outside evidence consistent with these predictions, especially Lerner, Sorensen, and Strömberg (2011), Bernstein (2011), Fang, Tian, and Tice (2010), and Asker, Farre-Mensa, and Ljungqvist (2011).

6 Short summary

- **Method.** A theoretical model of ownership structure, liquidity shocks, and insider trading under asymmetric information.
- **Core mechanism.** Private opacity creates tolerance for failure; public transparency creates short-termism.
- **Main result.** The equilibrium intensity of innovation is always at least as high under private ownership as under public ownership.
- **Ownership choice.** If innovative projects are more valuable, private ownership is preferred unless public capital/liquidity advantages dominate; if conventional projects are more valuable, public ownership is preferred.
- **Extension.** Illiquid private securities weaken the case for private ownership by adding a liquidity cost to the innovation advantage.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Public and private firms face systematically different innovation incentives because the information content of prices and trading opportunities differs across ownership forms.
- Private firms can innovate more because insiders can better absorb early failure.
- Public firms can innovate less because managers are tempted to choose projects that generate good early news and allow them to cash out.

7.2 Economic intuition

- Innovation is an activity with delayed payoff and high early failure risk.
- A system that punishes early failure less severely naturally encourages innovation.
- Private ownership does this by allowing bad-news sales to be pooled with liquidity-driven sales.
- Public ownership does the opposite by making good early outcomes especially valuable in the market.
- So the same market that improves transparency and liquidity can also create a bias toward safe, conventional ideas.

7.3 Takeaway

This paper's lasting insight is that **ownership structure is part of the firm's innovation technology**. Going public is not only a financing decision and going private is not only a control decision. Each ownership form changes how much failure the firm can tolerate and how strongly managers respond to short-run price signals. That is why private ownership tends to fit exploration and restructuring, whereas public ownership tends to fit harvesting and exploitation.